

CV

NAME: Richelle W. Kihoro

eRA COMMONS USER NAME (credential, e.g., agency login):

N/A POSITION TITLE: Research Scientist

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE	Start Date MM/YYYY	Completion Date MM/YYYY	FIELD OF STUDY
University of Nairobi, Kenya	BSc	05/2013	09/2017	Microbiology and Biotechnology
University of Nairobi, Kenya	MPH	09/2019	12/2023	Public Health Epidemiology and Biostatistics

• Personal Statement

I first became engaged in research during my undergraduate studies at the University of Nairobi, where I pursued a Bachelor of Science in Microbiology and Biotechnology. There I conducted a final year project supervised by Prof. Wolfgang Mukabana on identifying how economic creation activities in Lake Victoria facilitate the breeding of mosquitoes. This initial exposure to research laid a solid foundation for my subsequent endeavors. This led me to Kenya Medical Research Institute (KEMRI), where I volunteered as a Research Intern implementing the Early Infant Diagnosis (EID) program through conducting reverse transcription polymerase chain reaction (RT-PCR) assays for quantification of HIV on the Abbott m200 system and Roche CAP/CTM under principal investigator Prof. Matilu Mwau. Thereafter, I was recruited as a Research Scientist in KEMRI domiciled at Eastern and Southern Africa Centre of International Parasite Control (ESACIPAC). At ESACIPAC I am engaged in infectious and neglected tropical disease research on diseases such as soil-transmitted helminths, trachoma, lymphatic filariasis, schistosomiasis and COVID-19. At ESACIPAC I have conducted several serological analyses of dried blood spots (DBS) using the Bio-Plex Multiplex Immunoassay with the largest panel having over 46 different coupled antigens. I have also multiple conducted nucleic acid extraction and real time PCR experiments using the Step One Plus™ systems. At KEMRI I am also one of the participants selected for the KEMRI-JICA OMICS training under the Technical Cooperation Project for research capacity building of staff. I have been trained by Prof. Yoshito Fuji and Prof. Shingo Inoue on cell culture techniques, virus isolation techniques, expression and purification of recombinant protein in *E. coli* and bacterial identification by 16S rRNA gene sequencing. Overall, my extensive experience and training, coupled with my dedication to improving health outcomes, provide a strong foundation for my continued work in global health research. I am excited about the opportunity to further develop my skills and contribute to impactful research initiatives.

- **Positions, Scientific Appointments,**

and Honors Positions and Scientific

Appointments

2021 – Present	Research Scientist, Kenya Medical Research Institute
2023 – Present	Member, Royal Society of Tropical Medicine and Hygiene
2020 – 2020	Research Assistant, University of Nairobi
2018 – 2019	Research Intern, Kenya Medical Research Institute
2017 – 2017	Laboratory Technologist Intern, Centre for Health Solutions
2016 – 2016	Student on Industrial attachment, Kenya Medical Research Institute

Honors	
2023	President Fund Award, Royal Society of Tropical Medicine and Hygiene
2019	Scholarship for Master of Public Health tuition costs at University of Nairobi

C. Contributions to Science

Undergraduate Research: For my undergraduate capstone research project, I was supervised by Prof. Richard W. Mukabana at the University of Nairobi, whose research focuses on malaria. Under his guidance, I developed a dissertation exploring how economic activities around the Lake Victoria region contribute to the creation of mosquito breeding habitats, which in turn enhance malaria transmission. My thesis was titled, 'Man, Ecosystem Health, and Malaria: A Case Study of Mosquito Breeding Habitats on Rusinga Island in Western Kenya

Graduate Research: For my graduate studies I received a scholarship that covered my tuition costs to pursue a Master of Public Health at the University of Nairobi. My graduate research focus was on sexual and reproductive health and focused on determinants of use of long-acting reversible contraceptives among women in Nairobi county, Kenya. I graduated last year and I am currently developing a manuscript which I hope to submit by the end of the year to a peer-reviewed journal.

Professional Research: At ESACIPAC I am engaged in infectious and neglected tropical disease research on diseases such as soil-transmitted helminths, trachoma, lymphatic filariasis, schistosomiasis and COVID-19. At ESACIPAC I have conducted several serological analyses of dried blood spots (DBS) using the Bio-Plex Multiplex Immunoassay with the largest panel having over 46 different coupled antigens. I have also conducted multiple nucleic acid extraction and real time PCR experiments using the Step One Plus™ systems for xenomonitoring studies, COVID-19 and schistosomiasis. Additionally, I have conducted several GeneXpert tests that allow automation of sample purification, nucleic acid amplification and detection of target sequences using real-time PCR for detection of *Chlamydia trachomatis* in ocular swabs